MIL-D-55176C(EL)
18 October 1966
SUPERSEDING
MIL-D-55176B(EL)
3 April 1964

MILITARY SPECIFICATION

DATA, ELECTRONICS COMMAND ENGINEERING DOCUMENT CONTROL

1. SCOPE

1.1 This specification covers the entry of data into tabulating cards for use by the Electronics Command for control of Engineering Data.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

G-C-116	Cards; Tabulating

STANDARDS

MILITARY

MIL-STD-12	Abbreviations for use on Drawings and in Technical-Type Publications
MIL-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-155	Joint Photographic Type Designation System
MIL-STD-196	Joint Electronics Type Designation System
MIL-STD-804	Formats and Coding of Aperture, Copy and Tabulating Cards for Engineering Data Micro-reproduction System
MS 21319	Gage, Aperture Card, #201-1

PUBLICATIONS

FEDERAL

Cataloging Handbook H4-1 (Name to Code)	Federal Supply Code for Manufacturers

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer. Both the title and number or symbol should be stipulated when requesting copies.)

MISC

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3. REQUIREMENTS

3.1 Tabulating Cards

3.1.1 Card Stock. Tabulating card stock shall be Type 1 per G-C-116. Card Stock color shall be buff. Card size shall be in accordance with MIL-STD-804. The top and bottom edges of the card shall be thoroughly coated. This coating shall be composed of 25 percent vinyl acetate and 75 percent toluol.

3.1.2 Card Format. The tabulating cards shall be submitted on FMSC Form 32, dated 1 February 1962, "Listing Document (Card Code - 7)", see fig. 1, which shall be furnished by the contractor. The tabulating card manufacturer may imprint a marking for identification purposes along the bottom edge of the card, provided such imprint does not infringe upon the printed form. All printing, except the color stripes on the cards containing classified information (see 3.5) shall be in black ink.

3.1.3 Card Corner cut. All tabulating cards except those containing classified information shall have an upper left corner cut. The corner cut shall be 1/4 inch along the top edge of the card and 3/8 inch along the left edge of the card. Tabulating cards containing classified information shall have no corner cuts.

3.1.4 Interpretation. The information punched into the tabulating card shall be interpreted above the 12 punch location in solid line black characters at least .075 inch high in accordance with the interpreter bar assignments on the card format and in 3.4.13.1, 3.4.14.3, and 3.4.15.1.

3.2 Card punching code. The tabulating cards shall be punched using the card punching code in MIL-STD-804.

3.3 Decks of tabulating cards. The decks of tabulating cards that are the end product shall be furnished as separate decks. The contractor shall furnish one deck of tabulating cards for each listing document (see 6.3.1). The contractor shall furnish a separate, special deck of tabulating cards containing one duplicate of each "overflow" card prepared in accordance with 3.4.1. This is in addition to the "overflow" cards in the regular decks of tabulating cards. The contractor shall also furnish a separate, special deck of tabulating cards containing one duplicate of each header card in the regular decks of tabulating cards. This is in addition to the header card in each regular deck of tabulating cards.

3.3.1 Composition of a deck. The deck of tabulating cards for a listing document item shall contain a header card and one punched and interpreted tabulating card for each subsidiary document (see 6.3.2) cited on the listing document. Cards in a deck shall be furnished in the exact same sequence in which the subsidiary documents are cited on the listing document.

3.3.2 Identification of a deck. The deck of tabulating cards for a listing document shall be identified by the header card. The header card shall be a punched and interpreted tabulating card with the listing document identification entered in both the subsidiary document and the listing document fields in accordance with instructions in 3.4. In addition, the following information shall be entered into the header card in accordance with 3.4:

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(a) Type designation and model of item covered by the deck (See 3.4.14).

(b) Number of cards (included header card) in the deck (See 3.4.15).

3.4 Entry of data in cards. The codification and method of data entry into FMSC Tab Form 32 shall be in accordance with the following detailed instructions:

3.4.1 Overflow instructions. When a number (drawing, part, specification, etc.) is longer than the field in which it is to be entered, the number shall be entered starting in the left column of the field and continuing through the next to the last column of the field. An asterisk "*" shall be entered in the last column of the field to indicate that the number overflows the field. The remaining characters of the number shall be stamped, hand-printed, or machine printed in solid line black characters at least .075 inch high across the face of the card between the 11 and 12 punch locations with the first character directly beneath the interpreted asterisk*.

3.4.2 Prefix Letters field for subsidiary document (card columns 1 and 2). The associated list prefix letters of the subsidiary document in accordance with MIL-STD-100 shall be entered in this field.

3.4.3 Subsidiary Document Number field (card columns 3 thru 17). The number of the subsidiary document shall be entered in accordance with the following subparagraphs, which shall be applied to the sequence listed:

3.4.3.1 All numbers shall be left-hand justified.

3.4.3.2 Enter the number without change or modification in any manner except as shown in the following subparagraphs.

3.4.3.3 The six special characters: dash "-", slash "/", period or decimal ".", left parenthesis "(", right parenthesis ")", and asterisk "*", shall remain unchanged. In addition, the asterisk "*" shall be used to represent all other special characters.

3.4.3.4 Close up all spaces.

3.4.3.5 Do not enter the drawing size designator except as an element of a drawing number of which it is an integral part.

3.4.3.6 Do not enter the drawing revision symbol except as an element of a drawing number of which it is an integral part.

3.4.3.7 When an individual drawing is identified by multiple whole numbers the number entered shall be the first whole number in the series.

3.4.4 Code Identification Number field for subsidiary document (card columns 18 thru 22). The code identification number of a subsidiary document shall be entered in this field. The code identification number shall be determined using the following order of precedence.

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(a) Code Identification Number shown on the document.

(b) Code Identification Number listed in Cataloging Handbook H4-1 (Name to Code), Federal Supply Code for Manufacturers.

3.4.4.1 The Code Identification Number assigned to Electronics Command (formerly Signal Corps) drawings is "80063."

3.4.5 Revision Letter field for subsidiary document (card columns 23 and 24). The revision letter of the subsidiary document shall be entered in this field. If the revision is a single letter, it shall always be entered in card column 24. If the revision is identified by a date or by more than two characters in the revision symbol, an asterisk "*" shall be entered in card column 24. (See 3.4.1)

3.4.6 Subsidiary is a Listing Document code field (card column 25). If the subsidiary document is also a listing document as defined in 6.3.1, an 11 punch shall be entered in this field. If the subsidiary document is not a listing document, this field shall be left blank.

3.4.7 Prefix Letters field for listing document (card columns 26 and 27). The associated list prefix letters of a listing document in accordance with MIL-STD-100 shall be entered in this field.

3.4.8 Listing Document Number field (card columns 28 thru 42). The number of the listing document shall be entered in accordance with 3.4.3.

3.4.9 Code Identification Number field for listing document (card columns 43 thru 47). The code identification number for the design activity of a listing document shall be entered in this field in accordance with 3.4.4.

3.4.10 Revision Letter field for listing document (card columns 48 and 49). The revision letter of the listing document shall be entered in this field. If the revision is a single letter, it shall always be entered in card column 49.

3.4.10.1 If the revision of the listing document is identified by a date or by more than two characters in the revision symbol, an asterisk "*" shall be entered in card column 49. (See 3.4.1).

3.4.11 Card Code field (Card columns 50 and 51). The digit "7" shall be entered in card column 51. Card column 50 shall be left blank.

3.4.12 Security Classification field (Card column 52). The security classification of the subsidiary document or of the listing document, whichever is higher, shall be entered in this field in accordance with the code in MIL-STD-804.

3.4.13 Subsidiary Document Revised Field (card column 53). The number of equipment column used (see 3.10) shall be entered in this field.

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3.4.13.1 Card column 53 shall be interpreted across the face of the card between 0 and 11 punch locations by interpreter bar 23.

3.4.14 Title field (Card columns 54 thru 76). The title of the document whose number is entered in the SUBSIDIARY DOCUMENT NUMBER field shall be entered in these columns in accordance with the following order of precedence:

(a) Type designation in accordance with MIL-STD-196 or MIL-STD-155.

(b) Federal Stock number.

(c) Old Signal Corps stock number.

(d) Noun title.

3.4.14.1 As many words of the noun title as will fit in these 23 columns shall be entered. The title of a subsidiary document shall be entered as shown on the listing document. Abbreviations in accordance with MIL-STD-12 shall be used when necessary to permit entry of as much of the complete noun title as possible. If no title of the subsidiary document is shown on the listing document, these columns shall be left blank.

3.4.14.2 The first character shall always be in card column 54.

3.4.14.3 Card columns 54 thru 76 shall be interpreted across the face of the card between the 0 and 11 punch locations by interpreter bars 34 thru 56.

3.4.15 Listing Document Revised field (Card column 77) and card columns 78 thru 80. The number of cards (including header card) in the deck shall be entered in these columns of each header card. The last digit of the number of cards shall always be in card column 80 and non-significant zeros shall always be used to fill out the field. If the number of cards exceeds 9999, the letter A shall be used in card column 77 to indicate 10 (10,000 - 10,999), the letter B to indicate 11 (11,000 - 11,999), etc.

3.4.15.1 Card columns 77 thru 80 shall be interpreted across the face of the card between the 0 and 11 punch locations by interpreter bars 26 thru 29.

3.4.15.2 These columns shall be left blank in non-header cards.

3.5 Security identification. Tabulating cards containing classified information shall be marked in accordance with MIL-STD-804.

3.5.1 Classified information contained on tabulating cards shall be handled, controlled, transmitted and stored in accordance with applicable departmental security regulations and instructions, or if appropriated, in accordance with the Department of Defense

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Industrial Security Manual for Safeguarding Classified Information.

3.6 Card striping. The color and position of stripes used to visually identify tabulating cards containing classified information shall be in accordance with MIL-STD-804.

3.7 Verification of punched data. Each shipment of tabulating cards shall be accompanied by a brief statement describing the method used for verification of the accuracy of the data entered on the cards.

3.8 Workmanship. Tabulating cards shall be clean, un-bowed, free from nicks, folds, bent corners, torn corners, or other card damage that might affect punched card accounting machine processing. All printed and interpreted data shall be legible and unsmeared.

3.9 Numbering of Listing Documents. When it has been verified that a listing document does not show an identifying number for the listing document itself, a number shall be assigned from a block of special numbers provided by the procuring activity. This number shall be stamped or handwritten by the contractor in solid line black characters at least 3/16 inch high on the face of each sheet or page of the document, preferably near the lower right corner, so as to be right reading when the other data on the listing document is right reading. This number shall be entered on the card as the listing document number. The code identification number entered shall be that of the organization that prepared the listing document, not that of the procuring activity that furnished the block of special numbers.

3.10 ECOM Drawing Lists. Tabulating cards for an ECOM (or Signal Corps) Drawing and Data List, SELMS Form 1108 or FMSC Form 8880, shall be prepared in accordance with the following:

3.10.1 Figure 2 assigns a number to each of the seven equipment columns. These numbers are in ascending order from right to left.

3.10.2 The contractor shall work from only the highest numbered (left-most) column that has been used.

3.10.3 The equipment identification over this column shall be entered in the Title field of the header card (see 3.4.14). If the only identification in the column is a contract number, the contractor shall check the identification of the next lowest numbered column sequentially, until a type designation is reached. If none of the columns contain a type designation, the identification in the title block shall be used in accordance with the order of precedence in 3.4.14.

3.10.4 The number of this column, as shown on Figure 2, shall be entered in the Subsidiary Document Revised field (See 3. 4 .13).

3.10.5 A punched and interpreted tabulating card shall be furnished for each subsidiary document asterisked "*" in this column. Cards shall not be furnished for subsidiary documents not asterisked in this column.

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4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and test shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deem necessary to assure that supplies and services conform to prescribed requirements.

4.2. Classification of Inspection. Inspection shall be classified as follows:

(a) Acceptance inspection of items before preparation for delivery (see 4.3).

4.3 Acceptance inspection of items before preparation for delivery. The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified below. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government at its discretion may perform all or any part of the inspection specified herein, to verify the contractor's compliance with specified requirements. (The amount of such verification inspection will depend on the extend to which the contractor's inspection procedures and records insure that the product on contract meets the specified procurement inspection.)

4.3.1 This inspection (including sampling) shall conform to table 1 and the ordinary inspection procedures of MIL-STD-105.

TABLE I. - Inspection of Tabulating Cards

Inspection	Reqmt Para	Insp Para	AQL
Sequence Visual	3.3.1 3.1 3.2 3.4 3.5 3.6 3.7	4.5 4.6	.65% Major .65% Minor 2.5%

4.3.2 Verification. Before submitting tabulating cards to the Government for acceptance, the contractor shall verify, as a separate operation, the proper information has been correctly punched into each tabulating card.

4.4 Gage. Gage Number 201.1 shall conform to MS 21319. A tabulating card feed

4.4.1 Risk Plate. A common name for a [illegible]

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hopper weight weighing from seven to nine ounces shall be used to compress the cards in the gage. The cards shall be inserted, printed face down, so that the aperture area in the cards is positioned over the opening in the base of the gage.

4.5 Sequence inspection. A consecutive group of cards of the sample size "n" in accordance with MIL-STD-105 shall be selected without regard to quality, from a deck. The sequence of the cards in the sample shall be compared card by card with the sequence of the corresponding entries on the listing document, as determined by comparing the identification data on the first and last cards and the first and last entries on the listing document. If necessary in order to obtain the required size sample, the sample may include consecutive groups of cards from more that one deck of cards. Each time the identification data on the cards in the sample does not agree with the document identification data on the listing document, it shall be counted as one defect for the purposes of lot acceptance or rejection.

4.6 Visual Inspection of Tabulating Cards. The cards shall be inspected for the defects listed in table II.

TABLE II. - Classification of Visual Defects for Tabulating Cards

Classification	Defect
Major	Classified information not properly identified. Long edges of cards are visibly nicked in the areas exposed by gage number 201-1. Data not entered properly Data not interpreted properly Card is torn Card is folded or creased Format of card is incorrect Color of card is incorrect Color of stripe is incorrect Color of stripe is in wrong position
Minor	Corner of card is bent Card edge is visibly nicked in other than areas for major defects.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Tabulating cards shall be placed in suitable containers that will adequately protect the cards from physical damage. If overpacking is not used to provide the specified protection, each package shall be sufficiently rugged and moisture-resistant to protect the contents during domestic shipment. Each package shall contain a complete list of all decks of tabulating cards herein. Special decks of "overflow" cards and special decks of header cards shall be packaged and packed separately from regular decks."

5.2 Packing. Overpacking of the packages specified in 5.1 is permitted but not required.

5.3 Marking of Shipments.

5.3.1 Marking of packages. Each package shall be durably and legibly marked or labeled with the following information:

(a) Contractor's name

(b) Order number (File number of the contract)

(c) Decks of _____ cards (regular tabulating, overflow, or header - as applicable).

(d) "PACKAGE NO____ of____ PACKAGES"

(e) Other information as required by the contract

5.3.2 Marking of shipping cases. When shipping cases are used, they shall be marked in the same manner as specified in 5.3.1 for packages, except that item (d) shall be "CASE NO_____ OF____ CASES."

6. NOTES

6.1 Intended use. Tabulating cards furnished in accordance with this specification are intended for use as part of the engineering data control procedures of the U. S. Army Electronics Command.

6.2 Ordering data. Procurement documents should specify the following:

(1) Title, number and date of this specification and any amendment thereto.

(2) Acceptance at destination.

6.3 Definitions.

6.3.1 Listing document. A listing document is a tabulation of engineering documents (see 6.3.3) applicable to the item(s) to which the listing document applies. The following are examples of a listing document: Data List (DL) per MIL-STD-100; Index List (IL) per

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MIL-STD-100; Drawing and Data List; List of DRAWINGS. The following are not listing documents for the purposes of this specification: List of Material (LM); Parts Lists (PL) per MIL-STD-100.

6.3.2 Subsidiary document. A subsidiary document is an engineering document that is included in the tabulation on a listing document. A subsidiary document may be a detail drawing, an assembly drawing, a specification, a standard, or the listing document for a sub-item that forms a portion of the item(s) covered by the listing document on which the subsidiary document is cited. For example: Document X, the listing document for an IF strip, is a subsidiary document when it is cited on Document Y, the listing document for a radio receiver that includes that IF strip.

6.3.3 Engineering documents. "Engineering Documents" applies to specifications, drawings, sketches, lists, standards, pamphlets, reports and printed, typewritten, or other information, prepared by a design activity and relating to the design, procurement, manufacturer, test, or inspection of items or services.

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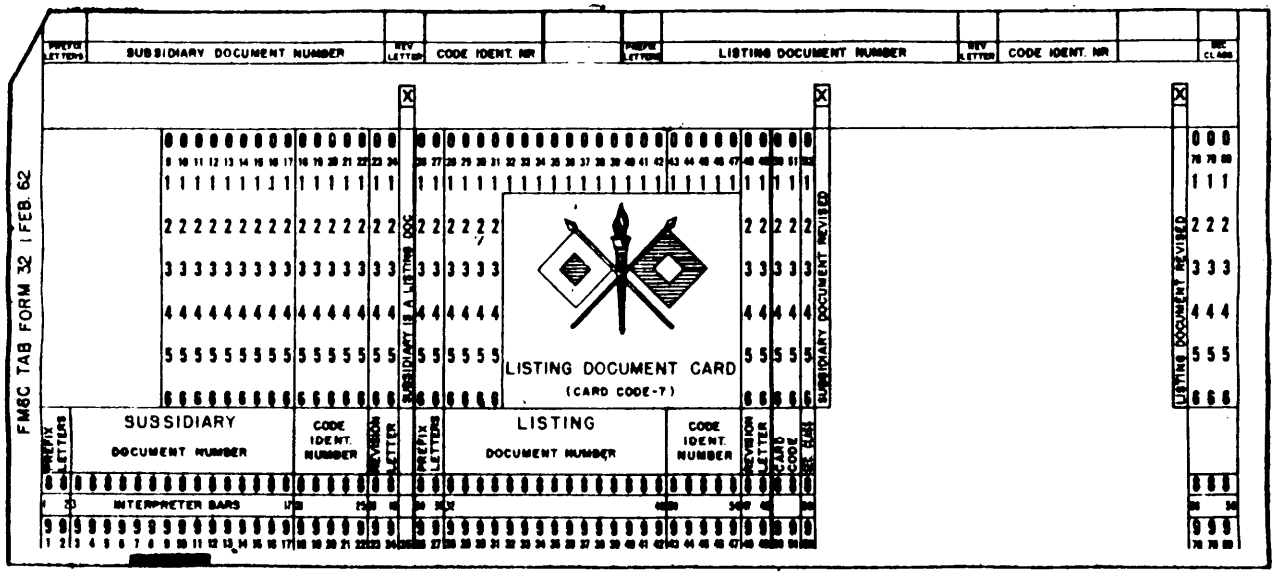


Figure 1. - LISTING DOCUMENT CARD

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EQUIPMENT	DEPARTMENT OF THE ARMY U.S. ARMY ELECTRONICS MATERIEL SUPPORT AGENCY FORT MONMOUTH NEW JERSEY **DRAWING AND DATA LIST** THIS DRAWING AND DRAWINGS AND PUBLICATIONS REFERENCED AND THEIR SUBSIDIARIES, IF ANY, CONSTITUTE ALL THE NECESSARY DATA RELATING TO DRAWINGS	SYM	REVISIONS	AUTH

	TITLE	NUMBER	REV. SYM.
7 6 5 4 3 2 1	Columns Numbers		

DRWN:	CHKD:	USED ON:	SC-DL-
APPD:	REVD:		SHEET-
DATE:			CODE IDENT 80063

TRIM LINE

Fig 2. Columns Numbers

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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